

Carlos Vega Rodríguez

Senior AI Architect & Full-Lifecycle Software Engineer

15+ years of software engineering, still actively shipping production code across React, Next.js, React Native, and AWS. Helps engineering organizations adopt AI in day-to-day work: shared conventions, review discipline, and repeatable AI-driven workflows that keep code consistent and quality high as teams scale. Works the full development cycle, from business goals to architecture and delivery.

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Experience

Senior AI Architect & Full-Lifecycle Software Engineer · 18Techs

SEP 2016 — PRESENT

Freelance / Contract · Remote · React Native · React Query · AWS · TypeScript · Next.js · Angular · React

- Implement AI-driven development-lifecycle practices into day-to-day processes.
- Deliver full-cycle product engineering across a portfolio of clients — mobile front ends, cloud back ends, and delivery.
- Standardize stacks, conventions, and CI/CD so teams ship consistently and resiliently.

// SELECTED CLIENT WORK

SOCIAL **munity**

Product, Software & AI Implementation Architect

Cross-platform community & social media app (iOS + Android) — media capture/upload, reactions, activity feeds, real-time chat, i18n, durable upload queues, and LLM-based content moderation, plus a Next.js admin panel and web invite system.

React Native · React Query · Next.js · AWS · LLM

AUDIT **Zinc.Tech**

AI-assisted Application Audit & Consultant

Independent, AI-assisted audit of a Java enterprise application (property-collections software), its REST API, and AWS architecture — a prioritized set of recommendations delivered to the client's engineering team.

- Reviewed the REST API and found roughly 40% of endpoints duplicated across roles; proposed an endpoint-by-endpoint consolidation strategy to remove dead code and realign business logic while keeping controllers live.
- Assessed AWS architecture and operational readiness — environment/region strategy, knowledge-transfer gaps, and the absence of security and load testing ahead of scaling.
- Evaluated client-side error handling and recommended error categorization, timeout and feedback loops, and clearer system-status indicators to eliminate stuck loading states and false-success outcomes.
- Delivered a prioritized, business-aligned remediation plan for the client's engineers.

Java · REST APIs · AWS · AI-assisted code analysis & audit

AD-TECH **QUAD**

Senior Cloud & DevOps Engineer

Complex, high-volume, always-available online ad delivery platform for a Stockholm-based ad-tech company — collaborated on system design and scaled cloud services for real-time ad delivery and reporting, operated in production and engineered for scalability and resiliency on AWS with Kubernetes.

AWS · Kubernetes · React · Angular

MOBILITY **Envoy**

Senior Mobile Application Engineer

EV car-sharing app ("Mobility as an Amenity") — app-based reservations, keyless vehicle access, and property-scoped fleet flows.

React Native · Mobile

Front-End Tech Lead · Informatech Costa Rica

JAN 2016 — SEP 2016

Costa Rica

- Led front-end development as tech lead.
- Re-platformed **Encore Capital Group's** consumer-facing sites.

Senior Software Engineer · Konrad Group

MAR 2014 — JAN 2016

San Pedro, San José, Costa Rica

- Delivered solutions for **Deloitte Digital**.

Senior Analyst · Accenture

NOV 2012 — MAR 2014

Costa Rica

- Delivered solutions to **SAY Media** and **DuPont**.

Web Developer & Designer · avVenta Worldwide

FEB 2012 — MAR 2014

Front-end development — CSS3 / HTML5 / JavaScript web pages & web apps

Web Developer · Samtec Interconnect (Intern)

AUG 2011 — DEC 2011

Costa Rica

Skills

LANGUAGES

TypeScript JavaScript SQL

FRONTEND WEB

React.js Next.js Angular HTML5 CSS3

MOBILE

React Native React Query Expo Redux Toolkit

BACKEND / CLOUD

AWS Node.js REST / GraphQL CI/CD

AI-AUGMENTED DEV

LLM workflows AI/LLM Claude Codex Cursor

PRACTICES

System Design Code Review & Standards CI/CD & Release State Management Offline-first
Testing i18n

AI in the Loop

AI integrated into the development process, with guardrails that protect quality and consistency at team scale.

- Implement standardized development processes and integrate AI into them across engineering organizations.
- Champion shared conventions, review discipline, and repeatable AI-driven workflows.
- Enforce guardrails against “AI slop” — AI-generated code meets the same bar as hand-written, via review, linting, and CI.
- Coach developers on AI pairing with Claude, Codex, and Cursor for architecture, refactors, and review.

Talks & Teaching

EDUCATION PARTNER

Mentor senior-year Computer Science design teams — application development (React / React Native), front-end architecture, and AI-assisted workflows

University of Central Florida (UCF)

Education

Bachelor's degree, Computer Software Engineering — Universidad Nacional de Costa Rica

2008 — 2011